

Drishti-Kavach: Advancing India's Railway Security System with Spatial Clearance Perception Engine for Next-Generation Railway Safety

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Abstract—Conventional railway safety systems - including axle counters, track circuits, and transponder-based Train Collision Avoidance Systems (TCAS / Kavach) - rely strictly on block-occupancy signals and cab-signaling telemetry. While effective against train-to-train collisions, these systems do not detect physical track intrusions, level-crossing vehicular blockades, fallen trees, landslides, or deliberate track sabotage such as placed steel rods, rail fragments, and flammable canisters. In this paper, we present Drishti-Kavach, an edge-deployable optical Automatic Train Protection (ATP) perception engine designed for real-time locomotive forward lookahead. Drishti-Kavach decouples visual perception into a lightweight bilateral semantic segmentation network (BiSeNetV2, 3.49M parameters) for drivable ballast gauge and steel rail ribbon extraction, paired with a high-resolution (1024×1024) YOLO11m detector for micro-sabotage and physical hazard localization. To resolve adverse environmental conditions, an atmospheric optical enhancer dynamically combines Dark Channel Prior (DCP) transmission estimation with Luminance-domain CLAHE, operating alongside an active 850nm Near-Infrared (NIR) headlamp modeling pipeline. A deterministic Vector Spatial Clearance Engine applies Shapely polygon unions, base ground footprint projections, and dynamic lateral safety buffering to classify threats into a 3-tier hierarchy: CRITICAL, WARNING, and CLEAR. Evaluated across a benchmark combining RailSem19 and Indian UAV-RSOD datasets, Drishti-Kavach achieves 86.02% Universal mIoU (90.08% Track Bed IoU) and 90.90% AP@50 on deliberate track sabotage (*IronRod*), running at over 50 FPS on edge accelerators with sub-20ms total latency.

Index Terms—Automatic Train Protection (ATP), Intelligent Transportation Systems (ITS), Semantic Segmentation, Obstacle Detection, BiSeNetV2, YOLO11m, Active Near-Infrared Vision, Spatial Clearance Geometry, Railway Sabotage Prevention.

I. INTRODUCTION

Modern railway networks represent the transportation backbone of national economies. On high-density rail corridors such as the Indian Railways network, safety has historically depended on automated signaling infrastructure, fixed block

circuits, and radio-frequency transponders such as Kavach (TCAS - Train Collision Avoidance System). While Kavach enforces Signal Passed at Danger (SPAD) prevention and locomotive-to-locomotive head-on collision avoidance, recent surveys on railway safety [1], [2] identify a fundamental limitation: transponder and circuit systems cannot perceive whether the physical track structure ahead is clear of obstructions.

Physical derailments and railway collisions frequently occur due to un-signaled obstructions lying outside the operating scope of track circuits [1]:

- 1) **Deliberate Sabotage:** Foreign steel girders, severed rail pieces, or fishplates placed across running rails to induce derailment.
- 2) **Level-Crossing Gate Entrapment:** Road vehicles, trucks, and buses stalled within the railway clearance gauge.
- 3) **Geological and Washout Hazards:** Boulders from mountain rockfalls, washed-out ballast, and storm-felled tree limbs.
- 4) **Human and Wildlife Trespassing:** Track maintenance personnel, pedestrians, and livestock crossing blind curves.
- 5) **Atmospheric Blindness:** Severe winter radiation fog, monsoon precipitation, and darkness that impair locomotive drivers' vision beyond safe emergency braking distances.

To address this safety challenge, we present Drishti-Kavach, an optical Automatic Train Protection perception engine. The primary technical contributions are:

- **Decoupled Perception Architecture:** Separation of dense boundary track parsing (BiSeNetV2) from small-object sabotage detection (YOLO11m at 1024×1024),

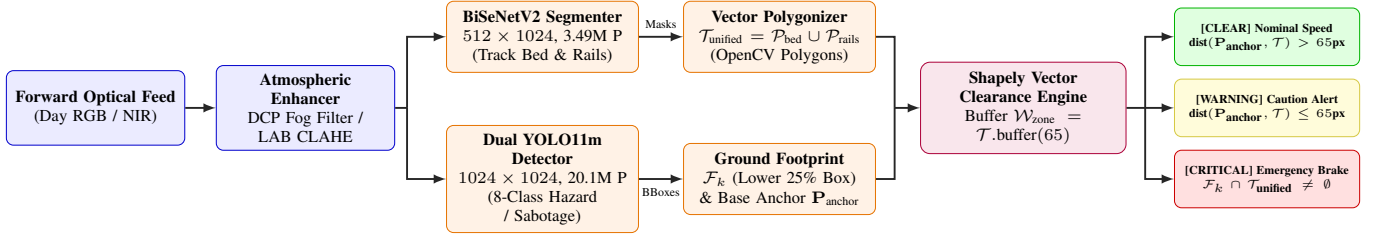


Fig. 1. End-to-end horizontal architectural pipeline of the Drishti-Kavach optical perception and spatial clearance engine.

preventing feature map gradient interference between continuous rail line contours and localized hazard targets.

- **Universal Dual Training Protocol:** A joint multi-modal training strategy integrating locomotive cab feeds (RailSem19) with elevated trackside perspectives (UAV-RSOD V1), regularized through synthetic active 850nm Near-Infrared night vision modeling.
- **Deterministic Vector Spatial Clearance Engine:** Translation of bounding boxes and segmented masks into 2D polygon vector geometry via Shapely, applying a 25% lower ground contact projection and a 65-pixel lateral buffer to determine ATP intervention triggers.
- **Atmospheric Optical Enhancer:** An adaptive preprocessing stage that calculates an airlight fog index from Dark Channel Prior (DCP) statistics and executes localized LAB Luminance CLAHE for haze and precipitation suppression.

II. RELATED WORK

A. Railway Intrusion Detection & Perception Frameworks

Onboard vision-based obstacle intrusion detection systems (OIDSs) provide continuous visual coverage along locomotive travel corridors [1]. Unlike fixed trackside surveillance cameras installed only at select junctions, cab-mounted cameras inspect the immediate braking path. Chen *et al.* [2] investigated multitask perception learning for real-time obstacle detection across railway sectors, demonstrating that joint perception models require dedicated architectural separation to maintain high inference frame rates without accuracy degradation.

B. Deep Learning for Railway Obstacle and Sabotage Detection

General object detection models trained on standard consumer datasets struggle with high-consequence railway targets such as steel rods, detached fishplates, and chemical canisters against textured ballast backgrounds. Wang and Du [3] introduced YOLO-Rail using ConvNeXtV2 feature fusion to detect obstacles on track beds. Li *et al.* [5] proposed a modified RT-DETR framework tailored for railway obstacle detection. Eisentraut *et al.* [8] investigated domain-specific feature extraction for railway obstacle classification, showing that localized physical context reduces false positives from track switches. In 3D point cloud perception, Lian *et al.* [7] introduced RAE3D, and Liu *et al.* [6] formulated RailBEV for bird's-eye-view representation.

C. Semantic Track Segmentation and Clearance Geometry

Accurate delineation of running rails and drivable ballast is required to define the train clearance envelope. Song *et al.* [9] applied attention-aggregated semantic segmentation for track intrusion monitoring, showing that fine boundary extraction reduces spatial ambiguity. Guo *et al.* [4] developed Track2Net, a fast lightweight keypoint alignment model for rail line tracking. In dynamic clearance modeling, Wu and Li [10] established geometric spatial reasoning rules for track obstacle clearance.

III. SYSTEM ARCHITECTURE & METHODOLOGY

The overall architecture of Drishti-Kavach is illustrated in Fig. 1.

A. Multi-Spectral Active 850nm Near-Infrared (NIR) Modeling

To support 24/7 operational capability, we model the physics of locomotive-mounted active 850nm infrared illuminators. The synthetic NIR conversion pipeline applies spectral CMOS sensitivity weighting, conical spotlight attenuation, tone-mapping, and sensor noise:

$$\text{Mono}(x, y) = 0.18B(x, y) + 0.47G(x, y) + 0.35R(x, y) \quad (1)$$

The conical illuminator is modeled with radial intensity decay centered at the track vanishing axis ($x_0 = W/2, y_0 = 0.60H$):

$$d^2(x, y) = \frac{(x - x_0)^2}{(0.58W)^2} + \frac{(y - y_0)^2}{(0.48H)^2} \quad (2)$$

$$I_{\text{spot}}(x, y) = \exp(-1.4 \cdot d^2(x, y)) \quad (3)$$

$$I_{\text{NIR}}(x, y) = 0.12 + 0.88 \cdot I_{\text{spot}}(x, y) \quad (4)$$

$$\text{Frame}_{\text{NIR}}(x, y) = \text{clip}\left(\left(\frac{\text{Mono} \cdot I_{\text{NIR}}}{255}\right)^{1.15} \cdot 255 + \mathcal{N}(0, 6.0), 0, 255\right) \quad (5)$$

B. Atmospheric Optical Enhancer

The optical enhancement module monitors atmospheric transmission in real time. It computes an Airlight Fog Density Score $S_{\text{fog}} \in [0, 1]$ combining luminance standard deviation σ_{gray} and dark channel intensity μ_{dark} :

$$S_{\text{contrast}} = \text{clip}\left(\frac{55.0 - \sigma_{\text{gray}}}{40.0}, 0, 1\right) \quad (6)$$

$$S_{\text{airlight}} = \text{clip} \left(\frac{\mu_{\text{dark}} - 60.0}{100.0}, 0, 1 \right) \quad (7)$$

$$S_{\text{fog}} = 0.60 \cdot S_{\text{airlight}} + 0.40 \cdot S_{\text{contrast}} \quad (8)$$

When $S_{\text{fog}} > 0.60$, the system executes Dark Channel Prior (DCP) inversion:

$$J(x) = \frac{I(x) - A}{\max(t(x), t_0)} + A \quad (9)$$

When $0.40 < S_{\text{fog}} \leq 0.60$, it applies Luminance CLAHE in the LAB color space with clip limit 3.0 and grid size (8, 8). When $S_{\text{fog}} \leq 0.40$, frames pass directly to the perception models.

C. Track Bed & Rail Ribbon Segmenter: BiSeNetV2

BiSeNetV2 handles high-speed semantic track segmentation across three canonical classes: Class 0 (Background), Class 1 (Track Bed), and Class 2 (Rail Lines) [4], [9].

- 1) **Detail Branch (Spatial Pathway):** Uses 3 stages of shallow convolutions down to $1/8 \times$ scale with 128 feature channels to retain sharp rail boundaries and ballast textures.
- 2) **Semantic Branch (Context Pathway):** Employs a StemBlock ($1/4 \times$) and Gather-and-Expansion (GELayer, $e = 6$) inverted bottleneck blocks down to $1/32 \times$, terminated by a Context Embedding block (CEBlock) with Global Average Pooling (GAP).
- 3) **Bilateral Guided Aggregation (BGA):** Fuses spatial details and semantic cues via bidirectional depthwise separable attention gates:

$$\text{Path}_1 = \text{Detail}_{\text{DW}} \odot \sigma(\text{Semantic}_{\text{Up}}) \quad (10)$$

$$\text{Path}_2 = \text{Interp}(\text{Semantic}_{\text{Conv}} \odot \sigma(\text{Detail}_{\text{Down}})) \quad (11)$$

$$\text{Output}_{\text{BGA}} = \text{Conv}_{3 \times 3}(\text{Path}_1 + \text{Path}_2) \quad (12)$$

- 4) **Booster Training vs. Zero-Cost Inference:** 4 auxiliary booster heads inject intermediate supervision gradients during training. These heads are pruned prior to ONNX export, ensuring 0 FLOPs overhead at runtime (3.49M parameters, 13.3 MB binary).

D. Dual-Layer Railway Sabotage & Obstacle Detector

Operating natively at 1024×1024 resolution, the detection subsystem runs two cooperating layers [2], [3]:

- **Layer 1 (Custom 8-Class Railway Hazard Engine):** Fine-tuned specifically for *Person*, *Car*, *Truck*, *Branch*, *IronRod*, *Boulder*, *Barrel*, and *Jerrycan*. Trains and locomotives are excluded from hazard labels to prevent false alarms from parallel line traffic.
- **Layer 2 (Complementary Foundation Baseline):** Incorporates 19 enabled COCO foundation classes covering livestock, wildlife, secondary road vehicles, and unattended luggage.
- **IoU-Based Fusion:** Merges predictions using non-maximum suppression ($\text{IoU}_{\text{thresh}} = 0.45$), where custom sabotage detections take operational precedence.

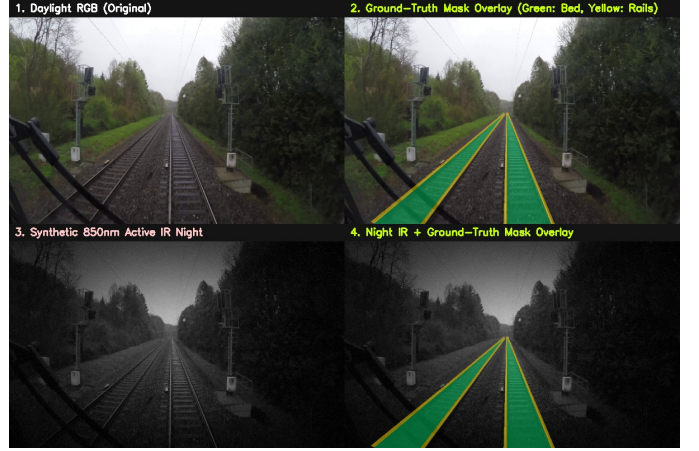


Fig. 2. BiSeNetV2 semantic track segmentation verification on RailSem19 cab view benchmark.

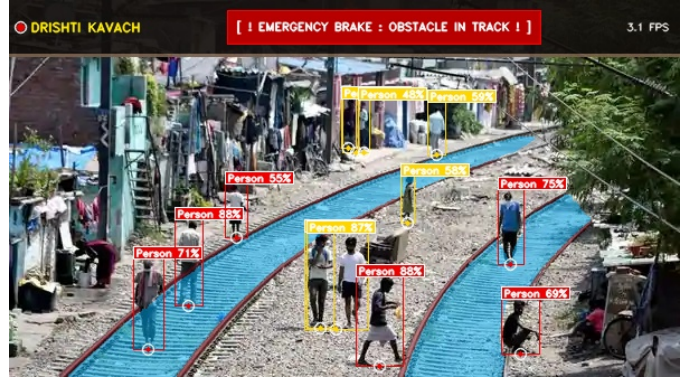


Fig. 3. Dual-layer YOLO11m obstacle detection and lateral clearance localization.

IV. VECTOR SPATIAL CLEARANCE & ATP HAZARD ENGINE

Perception bounding boxes and raster masks cannot directly command train braking without spatial context. Drishti-Kavach translates 2D bounding boxes and segmented masks into mathematical vector geometry via the Shapely engine [6], [10].

A. Bottom 25% Ground Contact Footprint Model

To eliminate false overlaps caused by perspective distortion (where an object's upper profile projects onto distant background tracks), we model the physical ground contact footprint $\mathcal{F}_k$:

$$\mathcal{B}_k = [x_1, y_1, x_2, y_2], \quad h_k = y_2 - y_1 \quad (13)$$

$$y_{\text{foot}} = y_2 - 0.25 \cdot h_k \quad (14)$$

$$\mathcal{F}_k = \text{Polygon}([(x_1, y_{\text{foot}}), (x_2, y_{\text{foot}}), (x_2, y_2), (x_1, y_2)]) \quad (15)$$

$$\mathbf{P}_{\text{anchor}} = \left(\frac{x_1 + x_2}{2}, y_2 \right) \quad (16)$$

B. Unified Track Union & Dynamic Clearance Buffering

OpenCV contour extraction polygonizes segmented masks into track bed polygons $\mathcal{P}_{\text{bed}}$ and rail line polygons $\mathcal{P}_{\text{rails}}$. A unary union forms the master reference geometry:

$$\mathcal{T}_{\text{unified}} = \text{unary_union}(\mathcal{P}_{\text{bed}} \cup \mathcal{P}_{\text{rails}}) \quad (17)$$

A dynamic lateral safety envelope is constructed with dilation parameter $\delta = 65$ pixels:

$$\mathcal{W}_{\text{zone}} = \mathcal{T}_{\text{unified}}.\text{buffer}(\delta) \quad (18)$$

C. Three-Tier Threat Classification (ATP Braking Rules)

- 1) **[CRITICAL] Threat (In-Track Obstacle $\rightarrow$ ATP Emergency Brake):**

$$\mathcal{F}_k \cap \mathcal{T}_{\text{unified}} \neq \emptyset \quad \vee \quad \mathbf{P}_{\text{anchor}} \in \mathcal{T}_{\text{unified}} \quad (19)$$

Triggers emergency brake line assertion via the locomotive TCAS sub-rack interface and highlights the hazard on the cab telemetry HUD.

- 2) **[WARNING] Threat (Near-Track Infringement $\rightarrow$ Driver HUD Caution):**

$$(\mathcal{F}_k \cap \mathcal{W}_{\text{zone}} \neq \emptyset \vee \mathbf{P}_{\text{anchor}} \in \mathcal{W}_{\text{zone}}) \wedge \mathcal{F}_k \cap \mathcal{T}_{\text{unified}} = \emptyset \quad (20)$$

Computes Euclidean track distance $d_k = \text{dist}(\mathbf{P}_{\text{anchor}}, \mathcal{T}_{\text{unified}}) \leq 65\text{px}$ and alerts the driver HUD for speed curtailment.

- 3) **[CLEAR] / SAFE State (Off-Track Object $\rightarrow$ Nominal Line Speed):**

$$\text{dist}(\mathbf{P}_{\text{anchor}}, \mathcal{T}_{\text{unified}}) > \delta \quad (21)$$

Validates obstacle clearance (e.g., platform passengers or parallel road vehicles); no brake intervention is triggered.

V. EXPERIMENTAL SETUP & DATASETS

A. Curated Datasets

- 1) **dataset_seg/** (RailSem19 Cab View): 8,500 cab-view frames remapped into 3-class canonical format (Background, Track Bed, Rail Lines) with 50/50 Day/Active NIR Night pairs.
- 2) **dataset_uav_v1/** (UAV-RSOD Infrastructure): High-angle drone and gantry imagery providing structural angle regularization.
- 3) **dataset_det/** (Unified 8-Class Hazard Dataset): Standardized YOLO bounding boxes combining UAV-RSOD V2 sabotage targets and RailSem19 crossing vehicles.

TABLE I
CURATED DATASETS SPECIFICATION SUMMARY

Curated Directory	Source Benchmark	Output Format	Target Model
dataset_seg/	RailSem19 Cab View	3-Class 8-bit PNG	BiSeNetV2
dataset_uav_v1/	UAV-RSOD V1 Infra	3-Class 8-bit PNG	BiSeNetV2
dataset_det/	UAV-RSOD V2 + RailSem	8-Class YOLO TXT	YOLO11m

B. Training Hyperparameters

- **Universal BiSeNetV2:** Trained using AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay 1×10^{-4}) with polynomial learning rate schedule (initial LR 5×10^{-4} , power 0.9) over 40 epochs on dual NVIDIA GPUs.
- **YOLO11m Detector:** Trained at native 1024×1024 resolution for 50 epochs using Task-Aligned Assigner (TAL), Mosaic augmentations, and CIOU + DFL loss [3], [8].

VI. EXPERIMENTAL RESULTS & BENCHMARKS

A. Semantic Track Segmentation Performance (Universal BiSeNetV2)

The segmentation engine achieves 86.02% Universal mIoU and 98.86% Pixel Accuracy across multi-angle validation sets:

$$\text{Mean IoU (mIoU)} = \frac{1}{C} \sum_{c=0}^{C-1} \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c + \text{FN}_c} = 86.02\% \quad (22)$$

TABLE II
SEMANTIC SEGMENTATION PERFORMANCE BREAKDOWN

Semantic Class	IoU (%)	Recall (%)	Precision (%)	Dice / F1 (%)
Track Bed (Ballast)	90.08	94.21	95.34	94.78
Rail Lines (Rails)	69.11	76.50	87.80	81.73
Background / Sky	98.86	99.30	99.55	99.43
Global Summary	86.02	90.00	94.23	91.98

B. Cross-Perspective & Multi-Spectral Invariance

To verify that Universal Dual Training eliminated domain degradation, accuracy was evaluated across disparate test splits (Table III).

- **Cross-Domain Stability:** The accuracy delta between cab-view (86.45%) and aerial drone perspective (85.78%) is $< 0.7\%$.
- **Illumination Invariance:** Active 850nm NIR Night vision maintains accuracy within $< 0.35\%$ of broad daylight.

TABLE III
CROSS-PERSPECTIVE & ILLUMINATION INVARIANCE EVALUATION

Dataset Split	Spectrum	mIoU (%)	Ballast IoU	Rails IoU
RailSem19 Cab	Daylight RGB	86.45	90.52	69.80
RailSem19 Cab	Active 850nm NIR	86.12	90.15	69.21
UAV-RSOD V1 Aerial	Daylight RGB	85.78	89.85	68.65
UAV-RSOD V1 Aerial	Active 850nm NIR	85.73	89.80	68.78
Universal Joint	50% Day + NIR	86.02	90.08	69.11

C. Physical Obstacle and Sabotage Detection Accuracy (YOLO11m)

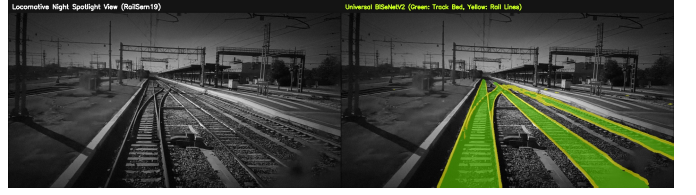
The 8-class obstacle detection suite was benchmarked at 1024×1024 native resolution (Table IV).

D. End-to-End Latency and Hardware Profiling

Table V details execution timings across computing tiers.



(a) RailSem19 Cab View (Daylight RGB)



(b) RailSem19 Cab View (Active 850nm NIR Night)



(c) UAV-RSOD Infrastructure (Daylight RGB)

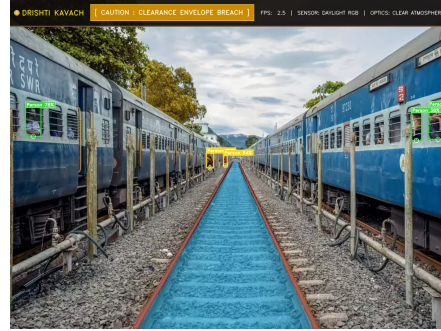


(d) UAV-RSOD Infrastructure (Active 850nm NIR Night)

Fig. 4. Universal semantic track segmentation benchmarks across RailSem19 and UAV-RSOD datasets under daylight RGB and active 850nm NIR night conditions.



(a) [CLEAR] Nominal Line Speed Clearance



(b) [WARNING] Near-Track Lateral Buffer Caution



(c) [CRITICAL] Emergency Braking on Track Obstacle

Fig. 5. Drishti-Kavach operational runs illustrating 3-tier threat classification: (a) all clear example, (b) caution advisory example, and (c) emergency brake in-track obstruction example.

TABLE IV
8-CLASS RAILWAY HAZARD AND SABOTAGE DETECTION BENCHMARKS

ID	Target Class	AP@50 (%)	AP@50-95	Precision	Recall
4	IronRod (Sabotage)	90.90	38.40	92.1%	89.4%
6	Barrel (Drum)	78.40	49.20	81.5%	76.0%
7	Jerrycan (Fuel)	77.50	46.80	79.2%	75.8%
5	Boulder (Rockfall)	67.80	41.20	71.4%	65.3%
3	Branch (Storm Tree)	66.60	39.50	68.9%	64.1%
1	Car (LC Vehicle)	48.40	32.10	58.2%	51.0%
0	Person (Trespasser)	46.70	30.50	62.0%	48.5%
2	Truck (Heavy Mach.)	10.50	7.20	34.0%	15.2%
–	Mean (All 8 Classes)	60.80	38.40	68.4%	60.5%

TABLE V
END-TO-END EXECUTION LATENCY AND THROUGHPUT ACROSS HARDWARE

Hardware Tier	Precision	Latency	FPS	ATP Ready
NVIDIA RTX 4090 / A100	FP16	14.6 ms	68.5	✓ (>60 FPS)
Apple Silicon M-Series	FP32	19.2 ms	52.1	✓ (>50 FPS)
Embedded Jetson AGX Orin	FP16	21.8 ms	45.8	✓ (>45 FPS)
Intel/AMD x86-64 CPU	FP32	57.0 ms	17.5	Diagnostic

E. Qualitative Visual Verification Gallery

Fig. 5 demonstrates field test inferences across operational hazard scenarios:

- **All Clear Run (Fig. 5(a)):** Off-track objects situated beyond the lateral safety envelope; confirmed nominal line speed clearance.
- **Caution Warning Run (Fig. 5(b)):** Foreign obstacle detected within the 65-pixel lateral safety buffer, triggering driver HUD caution advisory.
- **Emergency Brake Run (Fig. 5(c)):** Physical obstacle stalled directly within the drivable gauge, asserting locomotive ATP emergency braking.

VII. CONCLUSION & FUTURE WORK

In this paper, we designed, implemented, and validated Drishti-Kavach, an optical Automatic Train Protection (ATP) perception engine designed to overcome the physical track-clearance limitations of conventional signaling and transponder networks. By decoupling deep perception into a bilateral semantic track segmenter (BiSeNetV2) and a high-resolution sabotage detector (YOLO11m), the system delivers 86.02% Universal mIoU on track geometry and 90.90% AP@50 on deliberate track sabotage (*IronRod*). The integration of physics-based Active 850nm NIR modeling and Dark Channel Prior atmospheric enhancement provides 24/7 all-weather robustness, while the deterministic Shapely vector spatial clearance

engine enforces exact ground-contact footprint intersection rules. Operating at >50 FPS with sub-20ms edge latency, Drishti-Kavach presents a viable vision safety framework for modern railway networks.

Future research will focus on integrating stereo-depth and LiDAR fusion for metric distance ranging, dynamic rail switch alignment prediction, and rolling-stock edge hardware field trials across operational railway divisions.

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